Jurassic (Dinosaurs were dominant on the earth, fossil bird – ***Archaeopteryx***) and Cretaceous (extinction of toothed birds and dinosaurs, emergence of modern birds).

Cenozoic era (Age of mammals) is subdivided into two periods namely Tertiary and Quaternary. Tertiary period is characterized by abundant mammalian fauna. This period is subdivided into five epochs namely, Paleocene (placental mammals, Eocene (Monotremes except duck billed *Platypus* and *Echidna,* hoofed mammals and carnivores), Oligocene (higher placental mammals appeared), Miocene (origin of first man like apes) and Pliocene (origin of man from man like apes).Quaternary period witnessed decline of mammals and beginning of human social life.

The age of fossils can be determined using two methods namely, relative dating and absolute dating. **Relative dating** is used to determine a fossil by comparing it to similar rocks and fossils of known age. **Absolute dating** is used to determine the precise age of a fossil by using radiometric dating to measure the decay of isotopes.

6.3 Biological evolution

Formation of protobionts

Abiotically produced molecules can spontaneously self assemble into droplets that enclose a watery solution and maintain a chemical environment different from their surroundings. Scientists call these spheres as 'protobionts'. Liposomes are lipids in a solution that can self assemble into a lipid bilayer. Some of the proteins inside the liposomes acquired the properties of enzymes resulting in fast multiplication of molecules.

The coacervates with nucleoprotein and nutrients had a limiting surface membrane that had the characters of a virus or free living genes. Sub sequently number of genes united to form 'proto viruses' somewhat similar to present day viruses. Two major cell types that appeared during this time were significant. One form of the earliest cell contained clumps of nucleoproteins embedded in the cell substance. Such cells were similar to the Monera. They are considered as ancestral to the modern bacteria and blue green algae. The other form of earliest cells contained nucleoprotein clumps that condensed into a central mass surrounded by a thin membrane. This membrane separated nucleoproteins from the cell substances. Such cells were referred to as Protista. When the natural sources of food in the ocean declined in course of time the ancestors of Monera and Protista had to evolve different methods for food procurement. These may be summarized as parasitism, saprophytism, predator or animalism and chemosynthesis or photosynthesis. When the number of photosynthetic organisms increased there was an increase in the free O_2 in the sea and atmosphere.

$$CH_4+2O_2 \rightarrow CO_2 +2H_2O$$

$$4NH_3+3O_2 \rightarrow 2N_2+6H_2O$$

The atmospheric oxygen combined with methane and ammonia to form CO_2 and free nitrogen. The presence of the free O_2 brought about the evolution of aerobic respiration which could yield large amounts of energy by oxidation of food stuffs. Thus Prokaryotes and Eukaryotes evolved.

Experimental approach to the origin of life

Urey and Miller (1953), paved way for understanding the possible synthesis of organic compounds that led to the appearance of living organisms is depicted in the **Fig. 6.1.** In their experiment, a mixture of gases was allowed to circulate over electric discharge from an tungsten electrode. A small flask was kept boiling at 800°C and the steam emanating from it was made to mix with the mixture of gases (ammonia, methane and hydrogen) in the large chamber that was connected to the boiling water. The steam condensed to form

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